Hierarchical Multimodality Graph Reasoning for Remote Sensing Visual Question Answering

Han Zhang, Keming Wang, Laixian Zhang, Bingshu Wang, and Xuelong Li, *Fellow, IEEE*

***Abstract*— Remote sensing visual question answering (RSVQA) targets answering the questions about RS images in natural language form. RSVQA in real-world applications is always challenging, which may contain wide-field visual information and complicated queries. The current methods in RSVQA overlook the semantic hierarchy of visual and linguistic information and ignore the complex relations of multimodal instances. Thus, they severely suffer from vital deficiencies in comprehensively representing and associating the vision–language semantics. In this research, we design an innovative end-to-end model, named Hierarchical Multimodality Graph Reasoning (HMGR) network, which hierarchically learns multigranular vision–language joint representations, and interactively parses the heterogeneous multimodal relationships. Specifically, we design a hierarchical vision–language encoder (HVLE), which could simultaneously represent multiscale vision features and multilevel language features. Based on the representations, the vision–language semantic graphs are built, and the parallel multimodal graph relation reasoning is posed, which could explore the complex interaction patterns and implicit semantic relations of both intramodality and intermodality instances. Moreover, we raise a distinctive vision–question (VQ) feature fusion module for the collaboration of information at different semantic levels. Extensive experiments on three public large-scale datasets (RSVQA-LR, RSVQA-HRv1, and RSVQA-HRv2) demonstrate that our work is superior to the state-of-the-art results toward a mass of vision and query types.**

***Index Terms*— Hierarchical learning, parallel multimodality graph reasoning, remote sensing visual question answering (RSVQA).**

I. Introduction

REMOTE sensing imagery (RSI) contains a wealth of observation information on the surface of the earth. RSI data are playing an increasingly important role in a series of practical activities, such as vicinagearth security [1], emergency rescue, district management, precision agriculture, military reconnaissance, and so on. Due to the specialization of remote sensing tasks, the processing of visual information in RSI depends on experts in the specific domains. Various RSI interpretation techniques based on machine learning are experiencing rapid development, including scene classification [2], object detection [3], change detection [4], cross-modal retrieval [5], and so on.

Recently, multimodal techniques [6], [7] cooperatively analyzing the information from different perceptive modalities are intensively explored, such as visual question answering (VQA) [8], [9], image captioning [10], and image-to-text retrieval [5]. VQA aims to answer the questions about images in natural language form, enabling intelligent human–computer interaction to image understanding issues. Early VQA approaches [11], [12], [13], [14], [15] described fine-grained visual and language semantic information and adopted the linear fusion manner to integrate the vision and language representations. With the prevalence of attention mechanisms, a series of VQA methods [16], [17], [18] designed cross-modality attention schemes to discriminate multimodal objects and exploit the valuable information. Conventional VQA models focus on visual objects with meticulous semantics, not suitable for processing RSI data collected from the wide field of view.

VQA in remote sensing (RSVQA) has appeared in these years, providing the pathway to achieve indeterminate information in RSI for human beings. In the pioneering work of RSVQA [19], the authors built three public large-scale datasets and designed a basic approach for RSVQA. However, the model was too primitive and plain to learn premium fusion representations. To solve the problem, several models with different interaction strategies were proposed successively. Inspired by Fukui et al. [13], Ben et al. [14] and Chappuis et al. [20] separately applied multimodal compact bilinear pooling and tucker fusion to fuse features in different modalities. Taking advantages of attention mechanisms, a mutual attention network [21] was proposed to improve RSVQA performance. Considering RSVQA from the perspective of learning paradigms, Zhang et al. [22] and Zhang et al. [23] raised the easy-to-hard manner and question-driven manner. The research [24] modeled spatial relation reasoning in RSVQA. The work [25] utilized large pretrained models (CLIP [26] and BERT [27]) to encode and fuse information. The research [28] prompted large language models by transforming RSI into textual descriptions.

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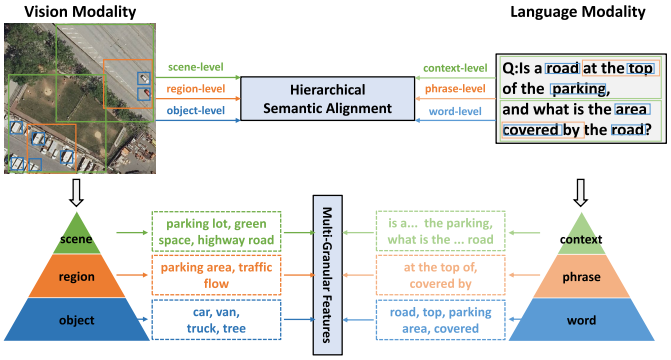


Fig. 1. Illustration of the hierarchical semantic alignment and multigranular feature representation in the vision–language RSVQA task.

Although previous methods [19], [20], [21], [22], [23], [25], [28] have focused on achieving semantic alignment between vision–language modalities to improve RSVQA performance, they still have two main shortcomings. On the one hand, existing methods **overlook the semantic diversities and hierarchies within modality correlation**, leading to coarse-grained multimodal semantic representations. As illustrated in Fig. 1, considering the large varying scales of objects or areas in RSVQA, extracting multigranular features from the vision and language modalities is crucial to achieving hierarchical semantic alignment. On the other hand, **they cannot effectively model complex interaction patterns and implicit semantic relations among different modalities**, causing models to fail to answer complicated RSVQA questions.

To conquer the challenges mentioned above, we propose the Hierarchical Multimodality Graph Reasoning network model, abbreviated as HMGR, which hierarchically learns the multigranular vision–language representation, and interactively parses the heterogeneous relationships of multimodal instances. An overview of the model is shown in Fig. 2. The overall contributions are summarized below.

1) We raise the hierarchical vision–language encoder (HVLE) and alignment structure to disentangle the multiscale semantic information, which can extract multigranular objects and areas to comprehensively understand the wide field of visual and textual contexts in remote sensing.
2) We design the parallel multimodal semantic graph interaction mechanism for vision–language relation reasoning, which helps associate the abundant relational knowledge from both intramodality and intermodality instances.
3) We propose a novel feature fusion method to aggregate hierarchical semantic features and collaborate on semantic relationships, via which the effective vision–language joint representation is optimized for RSVQA.
4) Extensive comparison and visualization experiments on the three benchmark datasets in RSVQA (LR, HRv1, and HRv2) are conducted, fully demonstrating that our model defeats the state-of-the-art competitors.

The rest of the article is structured as follows. Related works about VQA, RSVQA, and graph relation reasoning are introduced in Section II. Our method is detailed in Section III. Extensive experimental results and analysis are described in Section IV. Finally, the article is concluded in Section V.

II. Related Work

A. Visual Question Answering

The VQA task requires one to be capable of multimodal comprehension and reasoning between visual and textual information. The task attracts tremendous attention from both CV and NLP communities. In [8], [9], [29], [30], [31], and [32], the authors separately constructed six large-scale datasets, VQA1.0, VQA2.0, TDIUC, VQA-CP, CLEVR, and GQA. Besides, several articles [33], [34], [35] design various deep-learning architectures to generate answers. For instance, Noh et al. [33] proposed a dynamic parameter layer learned by questions in convolution neural networks. However, it remains a deficiency that these approaches often utilize monolithic vision representation with irrelevant and noisy information. Besides, inspired by the bilinear method, a sequence of multimodal bilinear pooling schemes [11], [12], [13], [14], [15] were proposed to describe mutual combinations of both modalities. However, these fusion methods result in an overall feature vector that cannot describe fine-grained semantic information. To solve the problem, [36], [37], [38], [39], [40], [41], [42] introduced diverse attention mechanisms into the VQA domain to find significant image areas and text contents.

B. Remote Sensing VQA

Compared with VQA studied in-depth, recent RSVQA is still in the initial stage. The pioneering work for RSVQA was introduced in [19]. In [19], to construct three datasets with different resolution levels, the authors designed an automatic question generation approach based on fixed templates and obtained remote sensing images from OpenStreetMap. On top of this, [21] built a public-facility-oriented dataset, called RSIVQA, with abundant scenes, including airports, ports, schools, industrial areas, and so on. In addition, some works [20] investigated the impact of bilinear fusion methods on RSVQA performance. However, previous models seldom consider the importance of learning modes. To fill these disadvantages, Yuan et al. [22] and Zhang et al. [23] proposed novel learning methods that enable models to solve RSVQA problems. Besides, [24] models spatial relation reasoning for RSVQA tasks. In addition, inspired by self-attention mechanisms and pretrained large transformer-based models, [25] leveraged BERT [27] and CLIP [26] to encode language and vision information, respectively. Apart from the above models, in [28], the researchers prompted large language models by few-shot learning.

C. Graph Relation Reasoning

The graph methods utilize a topological data structure to represent intricate relationships among many nodes with similar attributes. Recently, several VQA methods [43], [44], [45], [46], [47] based on graph relation reasoning have been proposed, demonstrating notable performance enhancements. For

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example, in [47], conditioned graph networks were employed to represent the relationships between objects. In the approach, visual objects and their connections were defined as graph nodes and edges, respectively. Besides, in [44], the graph attention mechanism was used to learn the semantic relations between the visual objects. In addition, in [43], significant pairwise relationships between objects were encoded through a multistep process, and the appearance features of the objects were then updated using these encoded pairwise relations. These methods all have a common drawback. The shortcoming stems from the lack of attention to the syntactic structure within the language modality and an exclusive emphasis on visual information. It poses a substantial challenge to the complex relation interpretation of various VQA tasks.

III. Method

A. RSVQA Problem Definition

Given a remote sensing image V and a matching question Q, the goal of the RSVQA task is to predict the answer a^* from all candidate answers $\mathcal{A}$. It can be formulated as

$$a^* = \arg\max_{a \in \mathcal{A}} F_\theta(a|Q, V) \tag{1}$$

where F_θ is the RSVQA model with learnable parameters θ and a is a candidate answer from $\mathcal{A}$.

B. Hierarchical Vision–Language Encoder

In this section, we elaborate on the HVLE for the RSVQA task. The overview of the module is shown in Fig. 3. Considering the scale changes of visual objects in remote sensing and semantic changes of textual description in NLP, we hierarchically represent the vision–language semantics at different levels.

1) Multiscale Vision Encoder: Given a remote sensing image V, as described in Fig. 1, we categorize visual semantics into object-level, region-level, and scene-level from small to large. It is known that the ResNet network effectively extracts visual features with distinct spatial information. Convolutional pooling operations could adjust the receptive field size to obtain visual features at different scales. We use pretrained ResNet34 and several convolutions and pooling operations to encode visual information with different spatial scales. Specifically, we first remove the last "pooling" and "flatten" layers of ResNet34 and fine-tune this part on the RSVQA training set. Then, the input images are fed into the encoder to obtain object-level feature maps $V^o \in \mathbb{R}^{L_o \times d_o}$. To obtain region-level and scene-level visual features $V^r \in \mathbb{R}^{L_r \times d_r}$, $V^s \in \mathbb{R}^{L_s \times d_s}$, we apply multiple steps of dynamic convolution and pooling operations to V^o. Besides, to effectively reduce gradient vanishing and explosion problems, improve model generalization ability, and speed up training, we apply the BatchNorm operation to the obtained visual representations. At this point, we have generated a set of feature vectors representing different-scale visual contents. The process of extracting visual features is formulated by

$$\begin{aligned} V^o &= \text{ResNet}(V) \\ V^r &= \text{Conv2D}\big(\text{MaxPool}\big(V^o\big)\big) \\ V^s &= \text{Conv2D}\big(\text{AvgPool}\big(V^r\big)\big) \\ \big\{\hat{V}^o, \hat{V}^r, \hat{V}^s\big\} &= \text{BatchNorm}\big(V^o, V^r, V^s\big) \\ \big\{\tilde{V}^o, \tilde{V}^r, \tilde{V}^s\big\} &= \text{ReLU}\big(W^o\hat{V}^o, W^r\hat{V}^r, W^s\hat{V}^s\big). \end{aligned} \tag{2}$$

Noting that the features are projected by three projection matrices $\{W^o, W^r, W^s\}$ to keep the same dimension d. Considering that the region-level representation is a mid-granular visual representation, max-pooling is used in extracting region-level information, such that some noise and redundant information at the object level could be discarded.

2) Multilevel Language Encoder: Aligning with visual instances, a multilevel language encoder is designed to achieve the abundant language information: "word-level," "phrase-level," and "sentence-level" language representations, denoted as $\{Q^w, Q^p, Q^c\}$. Thanks to the power to extract bidirectional textual semantics, the BERT model is employed in the proposed network. Given a question sequence Q, we utilize the Tokenizer module of the BERT model to map the input sequence into the textual features T. The Transformer module of the BERT model computes attention between words to optimize word features $Q^w \in \mathbb{R}^{L_w \times d_w}$, formulated as

$$\begin{aligned} T &= \text{Tokenizer}(Q) \\ Q^w &= \text{BERT}(T). \end{aligned} \tag{3}$$

Transformer-based BERT is a sequence-to-sequence model, and $Q^w = \{q_1^w, q_2^w, \cdots, q_m^w\}$ are made up of the feature vector of the single words by sequence. Three kinds of filters with varying sliding window sizes and receptive fields are designed to extract phrase-level information from textual sequences, including bigram (window size = 2), trigram (window size = 3), and quagram (window size = 4). The stride of these convolution filters is uniformly set to 1, and the length of output vectors aligns with the bigram filter by using the padding vectors. For a six-word input sentence, the output under the bigram filter contains three vectors, and the output under the trigram contains three vectors with one vector padding, while the output under the trigram contains three vectors with two vector padding. The max-pooling operation on features processed by the mixed filters to extract phrase-level language features $Q^p \in \mathbb{R}^{L_p \times d_p}$, that is,

$$Q^p = \text{MaxPool}(bi(Q^w), tr(Q^w), qu(Q^w)) \tag{4}$$

where bi, tr, and qu are, respectively, the abbreviations of bigram, trigram, and quagram filters. To obtain features of the same size, we used different padding settings. The multilayer gated recurrent unit (GRU) is leveraged to gather information on all of the words and form the context-level representation $Q^c \in \mathbb{R}^{L_c \times d_c}$

$$Q^c = \text{GRUCell}(Q^w). \tag{5}$$

Likewise, the LayerNorm operation is used to reduce the difficulty of model training, and three projection matrices $\{W^w, W^p, W^c\}$ are learned to ensure that all of the multilevel representations have the same dimension d, that is,

$$\begin{aligned} \big\{\hat{Q}^w, \hat{Q}^p, \hat{Q}^c\big\} &= \text{LayerNorm}\big(Q^w, Q^p, Q^c\big) \\ \big\{\tilde{Q}^w, \tilde{Q}^p, \tilde{Q}^c\big\} &= \tanh\big(W^w\hat{Q}^w, W^p\hat{Q}^p, W^c\hat{Q}^c\big) \end{aligned} \tag{6}$$

where $\tanh(\cdot)$ serves as the activation function.

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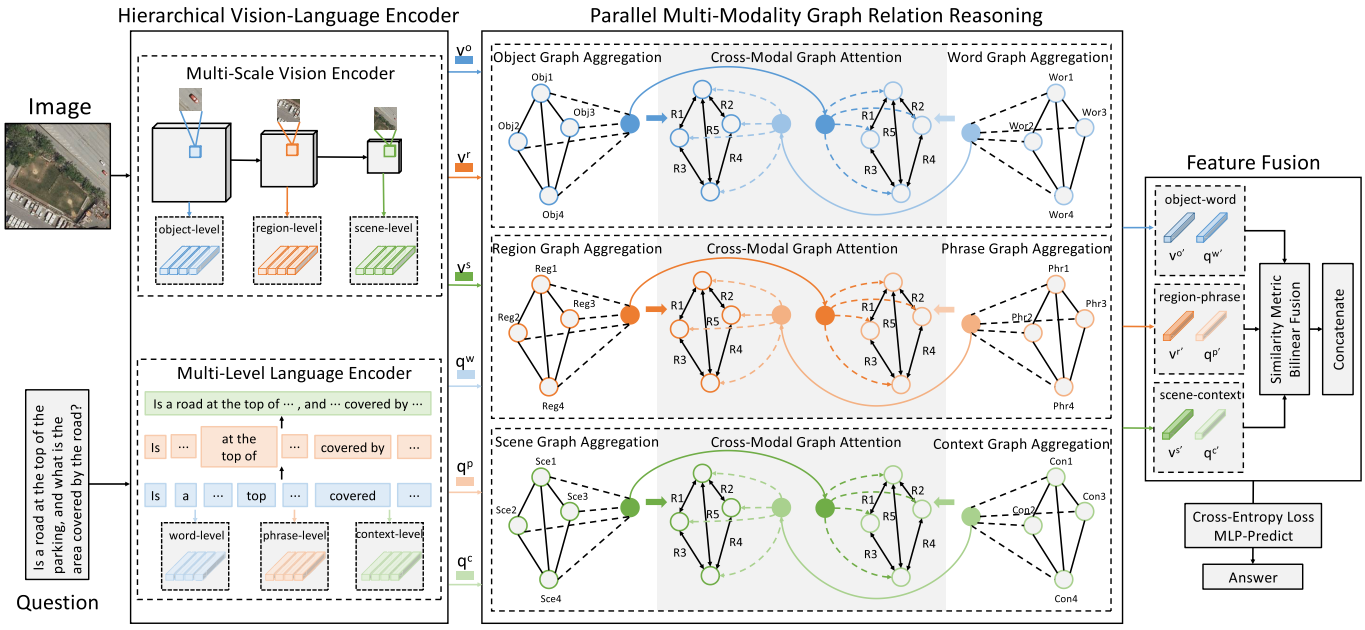


Fig. 2. Overview of the proposed hierarchical HMGR network, including three modules, that is, the HVLE, the PMGR, and the VQFS.

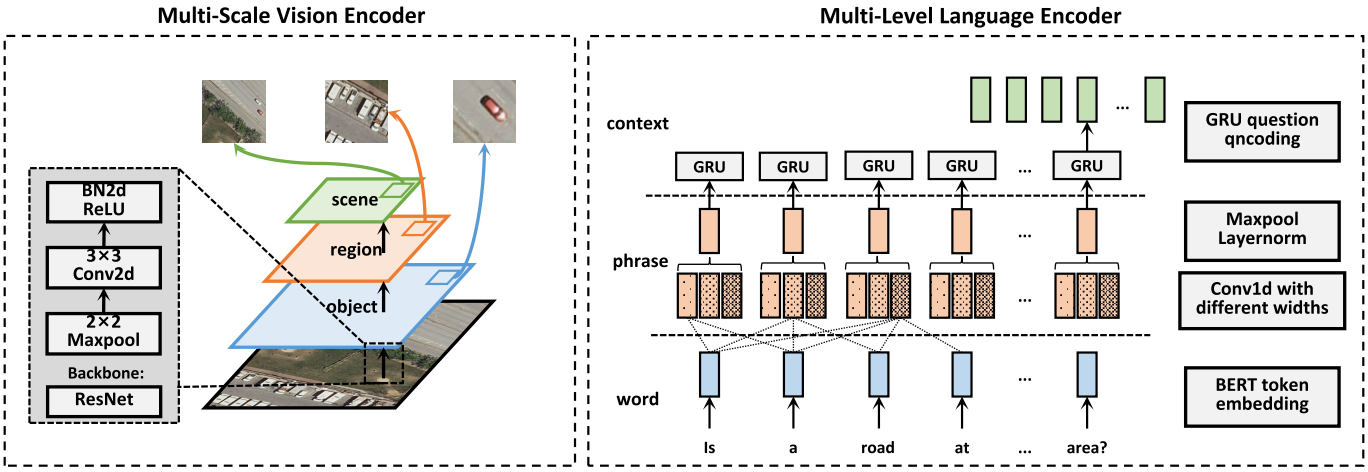


Fig. 3. Overview of the HVLE module, including the multiscale vision encoder and the multilevel language encoder. The visual modality is categorized into three semantic levels based on the scale of visual contents: object-level, region-level, and scene-level. Meanwhile, the language modality is also categorized into three semantic levels based on the organization logic of sentences: word-level, phrase-level, and context-level.

C. Parallel Multimodality Graph Relation Reasoning

With the hierarchical features above, we design the parallel multimodality graph relation reasoning module. The component overview is shown in Fig. 4. Considering the computational complexity, we perform cross-modal semantic alignment at three distinct levels: object-word, region-phrase, and scene-context, leveraging our previous experience as guidance. At each level, it begins by inferring intramodal relations and aggregating relevant information, and then the cross-modal graph attention is leveraged to enhance intermodal relationship modeling. The details are described as follows.

1) Graph Construction: For the given image V and the query Q, suppose $\mathcal{G}_{\mathcal{V}}$ and $\mathcal{G}_{\mathcal{Q}}$ denote the visual graph and language graph, respectively. Considering that the graphs in different modalities are constructed similarly, only visual modality is detailed in the following. Denote $h_{v_i} \in \mathbb{R}^d$ is the feature vector of the ith visual node v_i in the visual graph $\mathcal{G}_{\mathcal{V}}$. For each image, a fully connected semantic graph $\mathcal{G}_{\mathcal{V}} = \{\mathcal{N}_v, \mathcal{E}_v\}$ is initialized. $\mathcal{N}_v$ contains all vision nodes in the image, including object-level instances, region-level instances, and scene-level instances obtained before. All of the implicit relations among lots of represented instances are presupposed.

2) Intramodal Graph Attention Reasoning: We apply the graph attention mechanism to optimize the weights of the graph $\mathcal{G}_{\mathcal{V}}$, enabling the model to focus on significant relations and filter out noise relations, achieving semantic reasoning effects. The graph attention mechanism dynamically allocates higher weights to prominent semantic relations. Each node undergoes a shared bilinear transformation, defined by a weight matrix W_{θ_1}. The low-rank adaptation graph self-attention GraphSA among nodes to compute attention coefficients which indicate the importance of neighbor node

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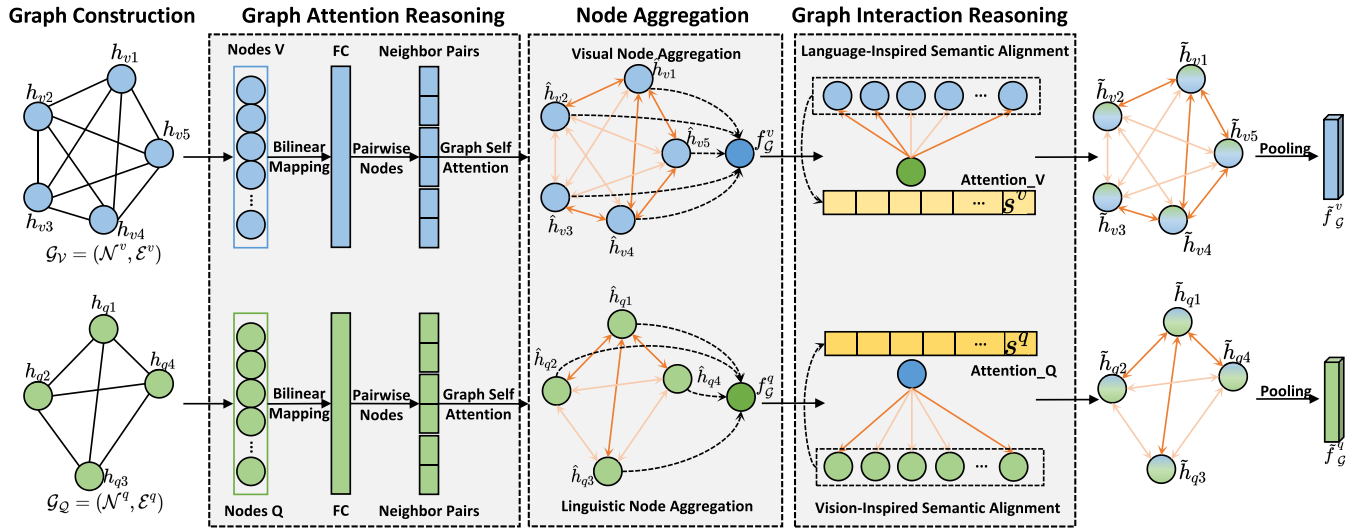


Fig. 4. Overview of the multimodality graph relation reasoning module at one pairwise level is illustrated. The graph attention reasoning module models potential relations of intramodal instances. The graph interaction reasoning module is designed to associate intermodality information. Two kinds of reasoning modules cooperatively capture multimodal semantic relations.

vector h_{v_j} to node vector h_{v_i}. Notably, the low-rank matrix on projection preserves the graph attention optimization capability while reducing computational overhead, achieving parameter efficiency. The process is formulated as

$$\alpha_{i,j}^{v} = \frac{h_{v_j}^{\top} W_{\theta_1} h_{v_i}}{\delta\sqrt{rd}} = \frac{h_{v_j}^{\top} \mathrm{BA} h_{v_i}}{\delta\sqrt{rd}} \tag{7}$$

where the original bilinear transformation matrix W_{θ_i} is decomposed into the product of two matrices $A \in \mathbb{R}^{r\times d}$ and $B \in \mathbb{R}^{d\times r}$, with the rank $r \ll d$. Instead of direct linear projection, we obtain d^2 parameters in W by computing $r \times d$ parameters in B and A, following a multiplier operation. Due to $r \ll d$ and $2dr \ll d^2$, the parameters to be optimized are decreased a lot by using the bilinear projection. So, it is more effective to compute the bilinear transformer than a normal linear transformation. The superscript $\top$ represents the transpose operation, δ is an adjustable constant used for scaling, and d is the node feature vector dimension. Besides, to facilitate the comparison of coefficients across different nodes, we use the softmax function to obtain normalized attention scores $\hat{\alpha}_{i,j}$ across all K neighboring nodes, formulated as

$$\hat{\alpha}_{i,j}^{v} = \mathrm{SoftMax}_j\left(\alpha_{i,j}^{v}\right) = \frac{\exp\left(\alpha_{i,j}^{v}\right)}{\sum_{k=1}^{K}\exp\left(\alpha_{i,k}^{v}\right)}. \tag{8}$$

Subsequently, we utilize normalized attention scores to propagate information of neighbor nodes. The node features $\hat{h}_{v_i}$ are updated after applying a linear transformation layer and nonlinear activation layer. To stabilize the learning process of graph self-attention, we extend the attention mechanism in a multihead manner, that is,

$$\hat{h}_{v_i} = \Big\Vert_{m=1}^{M} \sigma\left(\sum_{j=1}^{K} \hat{\alpha}_{i,j}^{vm} W_{\theta_2}^{m} h_{v_j}\right) + h_{v_i} \tag{9}$$

where M is the number of the multiheads. $\hat{\alpha}_{i,j}^{vm}$ are normalized attention scores computed by the mth attention mechanism, and $W_{\theta_2}^{m}$ is the corresponding input linear transformation weight matrix. In addition, we incorporate skip connection to prevent gradient vanishing. Then, the visual graph $\mathcal{G}_{\mathcal{V}}$ including the feature vectors of nodes and edges is updated as

$$\hat{G}_{\mathcal{V}} = \mathrm{ELU}\left(\left(\hat{\alpha}^{v} + I\right)\mathcal{G}_{\mathcal{V}}\right) \tag{10}$$

where $\mathrm{ELU}(\cdot)$ is the activation function, defined as

$$f(x) = \begin{cases} x, & x > 0 \\ e^{x} - 1, & x < 0. \end{cases} \tag{11}$$

ELU combines the characteristics of Sigmoid and ReLU functions. The linear portion on the right side enables ELU to alleviate gradient vanishing, while the left side enhances ELU's robustness to input variations or noise.

Furthermore, to obtain the representation of the entire graph, we utilize the AvgPool operation to aggregate information from all nodes, noted as

$$\hat{f}_{\mathcal{G}}^{v} = \mathrm{AvgPool}\left(\hat{G}_{\mathcal{V}}\right) \tag{12}$$

where $\hat{f}_{\mathcal{G}}^{v}$ is the graph feature of the given image.

3) Multimodal Graph Interaction Reasoning: To completely achieve visual and language interactions and explore the semantic relations between modalities, we propose this inference module. This module consists of two heuristic relationship attention computations: language-inspired attention and vision-inspired attention. Two attentions are independent and complementary to each other. Taking the language modality as an example, language-inspired attention injects textual semantic information into the visual graph. This involves guiding the adjustment of weights for individual nodes in the visual graph through language information. The goal is to achieve semantic alignment from vision to language. It can be formulated as

$$\begin{aligned} \tilde{h}_{v_i} &= \sigma\left(W_{\theta_3}\hat{h}_{v_i}\right) \odot \sigma\left(W_{\theta_4} f_{\mathcal{G}}^{q}\right) \\ s_{v_i} &= \mathrm{SoftMax}\left(W_{\theta_5}\tilde{h}_{v_i}\right) \\ \tilde{f}_{\mathcal{G}}^{v} &= \sigma\left(W_{\theta_6}\sum_{i} s_{v_i}\tilde{h}_{v_i}\right) + \hat{f}_{\mathcal{G}}^{v} \end{aligned} \tag{13}$$

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where $\tilde{h}_{v_i} \in \mathbb{R}^d$ is the visual graph node injected with the language guidance information, and $s_{v_i} \in \mathbb{R}^d$ represents the visual attention score of the ith node. Based on them, $\tilde{f}^v_{\mathcal{G}} \in \mathbb{R}^d$ absorbs the aligned vision–language relations into the representations of visual information and calculates the language-guided vision semantic graph. $W_{\theta_*} \in \mathbb{R}^{d\times d}$ is the trainable parameter matrices. Likewise, the vision-guided textual semantic graph is calculated. In this way, the dual relations in different modalities could be coupled, and the implicit relations could be perceived for reasoning complex queries.

4) Parallel Multimodality Graph Relation Reasoning: Based on the two modules mentioned above, the model computes the correlation within each modality and also measures the similarity among different modalities. It helps identify crucial nodes for answering different questions according to the specified images. To effectively leverage the multigranular features extract HVLE module, we devise the parallel multimodality graph relation reasoning network to improve computational efficiency. Considering three-level semantics are placed from coarse grit to fine grit, we establish the cross-modal semantic alignment scheme by aligning object-word level, region-phrase level, and scene-context level in the local-to-global order. For the object-word level, both visual and language features encapsulate highly concrete and localized semantic information. This is particularly beneficial for addressing questions that involve fine-grained, localized information, corresponding to different open-ended questions that can be effectively addressed. Notably, three semantic graphs are constructed and optimized by parallel computing.

D. VQ Fusion Module and Answer Prediction

In the VQA task, certain questions involve multiple semantic layers. However, in Section III-C, to reduce computational complexity, the model lacks these essential connections across the semantic layers, potentially limiting its capability. We present a new feature fusion approach to incorporate cross-level semantic connections while preserving the computation of reasoning. The main framework is shown in Fig. 5. According to the semantic pyramid structure, the module first performs similarity metrics among cross-level features in a bottom-up way. Then, it maps semantic information from the lower layers to the feature space of the upper layers. As a result, the model can achieve semantic interaction across levels with low computational costs. Taking the visual modality as an example, this process is formulated as follows:

$$\begin{aligned} \lambda &= \mathrm{CosSim}\left(f^b, f^u\right) \\ f^{u'} &= f^u + \lambda f^b \end{aligned} \tag{14}$$

where λ is the cosine similarity between the bottom-layer feature f^b and the up-layer feature f^u. The sum operation fuses the two representations in the vision modality. The language modality is processed similarly.

In addition, the bilinear fusion method is adopted to effectively interact between different modal features. The attention concatenation method is utilized to capture the optimal representation across multiple modalities noted as

$$H = \mathrm{ELU}\left(W_{\theta_6} F^q \odot W_{\theta_7} F^v\right)$$

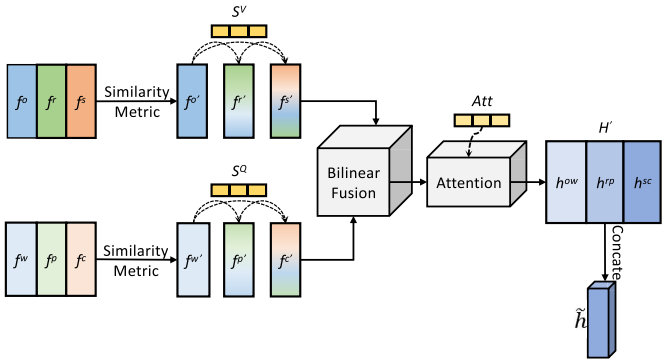


Fig. 5. Illustration of the VQ fusion scheme. The module obtains the compact and subtle vision–language joint representation from multiple semantic levels.

$$\begin{aligned} \alpha &= \mathrm{SoftMax}\left(W_{\theta_8} H\right) \\ \tilde{h} &= \mathrm{Concat}(\mathrm{Reshape}(\alpha H)) \end{aligned} \tag{15}$$

where $\{W_{\theta_6}, W_{\theta_7}, W_{\theta_8}\}$ are trainable weights. H is the interaction feature matrix from the visual representation matrix $F^v = [f^o, f^r, f^s]$ and the linguistics representation matrix $F^q = [f^w, f^p, f^c]$. Besides, α is the attention weights and $\tilde{h}$ is the final vision–language joint representations.

The final joint representation is fed into the two-layer MLP classifier to predict corresponding answers. And we use the cross-entropy loss function to optimize the parameters. It is formulated as

$$\mathcal{L} = -\sum_{i=1}^{n} y_i \log\left(\mathrm{MLP}\left(\tilde{h}\right)\right) \tag{16}$$

where y_i is the label of the answer to the ith image–question pair, and $\mathrm{MLP}(\tilde{h})$ represents the corresponding probability.

IV. Experiments

A. Experimental Settings

1) Datasets: To verify the validity of the model, we evaluate the model on three public large-scale datasets (LR, HRv1, and HRv2). These three datasets are released in the work [19]. The LR dataset contains 772 images from OpenStreetMap, taken by the Sentinel 2 satellite at 10-m resolution. Each image has 256 × 256 pixels. In addition, there are 77 232 question–answer pairs in this dataset. In specific, the authors categorize the pairs into count, presence, comparison, and rural/urban, with percentages of 29.78%, 29.63%, 39.59%, and 1.00%, respectively. Then, 77.8%, 11.1%, and 11.1% of the original data are divided into the training, validation, and test sets. As for the HR(v1 and v2) datasets, they share the same training and validation sets but have different test sets. In specific, they use 15-cm resolution aerial images extracted from the USGS high-resolution orthoimagery. They contain a total of 10 659 images with a pixel size of 512 × 512 and 1 066 316 question–answer pairs. Unlike the LR dataset, the authors divide the question–answer pairs into count, presence, comparison, and area which account for 26.04%, 26.10%, 33.18%, and 14.68%, separately. Then, 61.5%, 11.2%, 20.5%, and 6.8% of the original data are split into the training, validation, test-v1, and test-v2 sets, respectively. The HRv1 dataset consists of training, validation, and test-v1 subsets.

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Fig. 6. Visualization of qualitative samples on the three datasets (LR, HRv1, and HRv2 datasets) by rows, respectively. The first column consists of images, questions (Q), GT, and predicted answers (A). The second to fourth columns represent attention distributions at different SHs. Note that the visual attention scores increase sequentially from purple to red and the attention scores rise in sequence from gray to black.

2) Competitors: The compared methods contain five representative and state-of-the-art algorithms in the RSVQA task, as below.

1) *RSVQA [19]:* This pioneering work is the baseline in the field. It employs ResNet to learn visual representations, utilizes SkipThought to learn textual representations, and combines features using element-wise multiplication.
2) *E2H [22]:* The method designs an easy-to-hard manner to guide the visual information extracting guided by language information. It supervises model training and leverages scaled dot-product attention to enhance the interaction between images and language.
3) *BM [25]:* A novel transformer-based model that conducts intermodal interaction and reasoning between language and vision branches separately.
4) *SHRN [24]:* The work focuses on modeling the spatial relationships of visual objects, capturing intermodal relationships using scaled dot-product attention.
5) *MMFT [48]:* The approach leverages the global attention modeling capability of the transformer model

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for joint reasoning between visual and language modalities.

3) Implementation Details: In the HVLE, for the language encoder, we set the word-level $\tilde{Q}^w$, phrase-level $\tilde{Q}^p$, and context-level $\tilde{Q}^c$ feature vector dimensions to 768. For the vision encoder, the feature vector dimensions of the object-level $\tilde{V}^o$, region-level $\tilde{V}^r$, and scene-level $\tilde{V}^s$ are set to 512. Then, in the intramodal graph attention reasoning part, the dimensions of the linguistic graph representation $f_G^{Q'}$ and the visual graph representation $f_G^{V'}$ are set to 768 and 512, respectively. After this, for the cross-modal graph interaction reasoning part, we set the linguistic and visual graph representation dimensions $\{f_G^{Q''}, f_G^{V''}\}$ to 1200. Besides, the vision–question (VQ) interaction feature vector $\{\tilde{h}, h^{ow}, h^{rp}, h^{sc}\}$ dimensions are set to 1200, consistent with the above module. And the dimensions of kth textual representation q_L^k, visual representation v_L^k, and unified representation f_L^k are set to 1200. In addition, the BERT and ResNet models are initialized by pretraining weights and fine-tuned over the RSVQA datasets. We use the Adam optimizer for all experiments to progressively update the parameters and set the learning rate to $1e^{-5}$. To prevent overfitting, we introduce the L2 regularization term and set its weight as $1e^{-6}$. On the other hand, we use dropout with a probability of 0.1. Furthermore, accuracy is used as the evaluation metric. For the LR dataset, the batch size, length of the sentence, and number of the learning epoch are set to 32, 28, and 20, respectively. As for the HRv1 and HRv2 datasets, the batch size, embedding length, and learning epoch are set to 32, 35, and 10 correspondingly. To avoid errors and mistakes, we conduct three independent experiments to calculate the mean and variance.

B. Comparisons on RSVQA Benchmark Datasets

In this section, we compare our proposed HMGR model with four other RSVQA works on three public large-scale datasets mentioned above.

As the experimental results presented in Table I, our model outperforms the state-of-the-art results on all types of questions. Specifically, the model accuracy was 94.00% on the rural/urban type, 91.35% on the presence type, 74.64% on the count type, and 91.70% on the comparison type. Generally, compared to other methods, our model enhances average accuracy in the range from 0.58% to 6.43%, and overall accuracy in the range from 0.74% to 7.51%. The result demonstrates that our model can effectively answer questions about RS images with low resolution. In addition, our model exhibits small variances across all metrics, indicating good stability.

Compared with other well-performing RSVQA models, our proposed method achieves competitive results on most question categories of the RSVQA-HRv1 dataset presented in Table II. Specifically, the model accuracy was 86.54% on the area type, 92.09% on the presence type, 70.24% on the count type, and 91.99% on the comparison type. Our model enhances average accuracy in the range from 0.08% to 2.09%, and overall accuracy in the range from 0.14% to 2.30%. The result shows that our work can solve RSVQA problems with high resolution. Besides, our model shows small variances across different types of problems, which indicates good robustness.

As presented in Table III, our work outperforms the best results on different question types of the RSVQA-HRv2 dataset. In detail, the model accuracy was 86.24% on the area type, 93.04% on the presence type, 67.31% on the count type, and 94.04% on the comparison type. Our work improves average accuracy in the range from 4.39% to 7.65%, and overall accuracy in the range from 4.27% to 7.41%.

On top of this, we perform a qualitative analysis by visualizing the cross-modal graph inference processes at multiple levels, as shown in Fig. 6. The first row represents the sample from the LR dataset, the second row represents the sample from the HRv1 dataset, and the third row represents the sample from the HRv2 dataset. The first column contains the input RS images, questions, ground truths (GTs), and prediction answers from the three datasets. The second to fourth columns present the attention distribution at three semantic hierarchies (SH)—word-object, phrase-region, and context-scene, which indicates the textual keywords and the critical image regions that the model focuses on during the graph reasoning. For the visual modality, attention scores increase sequentially from purple to red. For the language modality, attention scores rise in sequence from blue to black. And to some extent, these attention distribution maps provide interpretability for the answer results.

C. Hyperparameter Sensitivity Analysis

To further investigate the effects of the hyperparameters—training data size, batch size, and head number on the model, we carry out two following ablation experiments, and the results are displayed in Tables IV–VI. In Table IV, we compare the performance of different batch sizes on the LR dataset and we observe minimal differences in these results, indicating that our model is not sensitive to the batch size parameter. In Table V, we use 25%, 50%, 75%, and 100% of LR data for model training. We can find that the accuracy of most question types gradually rises as the amount of data increases. In addition, we also find that the improvement of the model tends to slow down when the training size reaches more than 50%, which somehow indicates that the model does not depend on large-scale data. Table VI studies the impact of the number of attention heads. It is observed when the number of attention heads is less than 4, the model performance gradually improves with an increase in the number of attention heads. When the number of attention heads is greater than 4, the model performance stabilizes, indicating convergence.

D. Ablation Study

Our complete VQA model incorporates various crucial submodules, and there are significant hyperparameters that warrant discussion and validation.

1) Effectiveness of Submodules: In this section, we initially conducted several ablation studies using different versions of HMGR to assess the impact of each submodule on the final predictive performance.

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TABLE I
COMPARISON PERFORMANCE [MEAN(STD)] ON THE LR DATASET. THE BEST RESULTS ARE IN BOLD

Types	RSVQA-LR					
	RSVQA [20]	E2H [23]	MMFT [49]	BM [26]	SHRN [25]	**ours**
Count	67.01% (0.59%)	69.22% (0.33%)	69.36% (–)	72.22% (0.57%)	73.87% (0.22%)	**74.64% (0.12%)**
Presence	87.46% (0.06%)	90.66% (0.24%)	89.00% (–)	91.06% (0.17%)	91.03% (0.13%)	**91.35% (0.24%)**
Comparison	81.50% (0.03%)	87.49% (0.10%)	83.46% (–)	91.16% (0.09%)	90.48% (0.05%)	**91.70% (0.01%)**
Rural/Urban	90.00% (1.41%)	91.67% (1.53%)	88.00% (–)	92.66% (1.52%)	94.00% (0.87%)	**94.00% (0.04%)**
Average Accuracy	81.49% (0.49%)	84.76% (0.35%)	82.45% (–)	86.78% (0.28%)	87.34% (0.13%)	**87.92% (0.11%)**
Overall Accuracy	79.08% (0.20%)	83.09% (0.15%)	80.99% (–)	85.56% (0.16%)	85.85% (0.28%)	**86.59% (0.17%)**

TABLE II
COMPARISON PERFORMANCE [MEAN(STD)] ON THE HRV1 DATASET. THE BEST RESULTS ARE IN BOLD

Types	RSVQA-HRv1				
	RSVQA [20]	E2H [23]	BM [26]	SHRN [25]	**HMGR(ours)**
Count	68.63% (0.11%)	69.06% (0.13%)	69.80% (0.09%)	70.04% (0.15%)	**70.24% (0.11%)**
Presence	90.43% (0.04%)	91.39% (0.15%)	92.03% (0.08%)	**92.45% (0.11%)**	92.09% (0.06%)
Comparison	88.19% (0.08%)	89.75% (0.12%)	91.83% (0.00%)	91.68% (0.09%)	**91.99% (0.22%)**
Area	85.24% (0.05%)	85.92% (0.19%)	86.27% (0.05%)	86.35% (0.13%)	**86.54% (0.13%)**
Average Accuracy	83.12% (0.03%)	83.97% (0.06%)	84.98% (0.05%)	85.13% (0.08%)	**85.21% (0.10%)**
Overall Accuracy	83.23% (0.02%)	84.16% (0.05%)	85.30% (0.05%)	85.39% (0.05%)	**85.53% (0.21%)**

TABLE III
COMPARISON PERFORMANCE [MEAN(STD)] ON THE HRV2 DATASET. THE BEST RESULTS ARE IN BOLD

Types	RSVQA-HRv2				
	RSVQA [20]	E2H [23]	BM [26]	SHRN [25]	**HMGR(ours)**
Count	61.47% (0.08%)	61.95% (0.08%)	63.06% (0.11%)	63.42% (0.14%)	**67.31% (0.12%)**
Presence	86.26% (0.47%)	87.97% (0.06%)	89.37% (0.21%)	89.81% (0.27%)	**93.04% (0.03%)**
Comparison	85.94% (0.12%)	87.68% (0.23%)	89.62% (0.29%)	89.44% (0.23%)	**94.01% (0.05%)**
Area	76.33% (0.50%)	78.62% (0.23%)	80.12% (0.39%)	80.37% (0.16%)	**86.24% (0.22%)**
Average Accuracy	77.50% (0.29%)	79.06% (0.15%)	80.54% (0.16%)	80.76% (0.21%)	**85.15% (0.12%)**
Overall Accuracy	78.23% (0.25%)	79.29% (0.15%)	81.23% (0.15%)	81.37% (0.19%)	**85.64% (0.03%)**

1) *−HVLE:* We remove the hierarchical vision–language module from the proposed HMGR model. Instead, the model can only extract features from the final layer of the CNNs and RNNs.
2) *−Parallel multimodal graph relation reasoning (PMGR):* We remove the parallel multimodality graph reasoning module from the proposed HMGR model. As a substitute, we directly input the obtained multiscale features into the feature fusion module.
3) *−Vision question feature fusion (VQFS):* We remove the VQ feature fusion part from the proposed HMGR model. Alternatively, we use a simple Hadamard product to fuse various features and concatenate them directly.
4) *+HVLE:* We only add the hierarchical vision–language module into the baseline algorithm [19], and the same as +PMGR and +VQFS.

We respectively conduct ablation studies on the above three datasets, and the results are shown in Table IV. On the one hand, compared with the HMGR without the HVLE, the OA and AA of the full model increase by 0.91% and 0.35% on the LR dataset, by 0.19% and 0.22% on the HR(v1) dataset and 0.27% and 0.29% on the HR(v2) dataset. Besides,

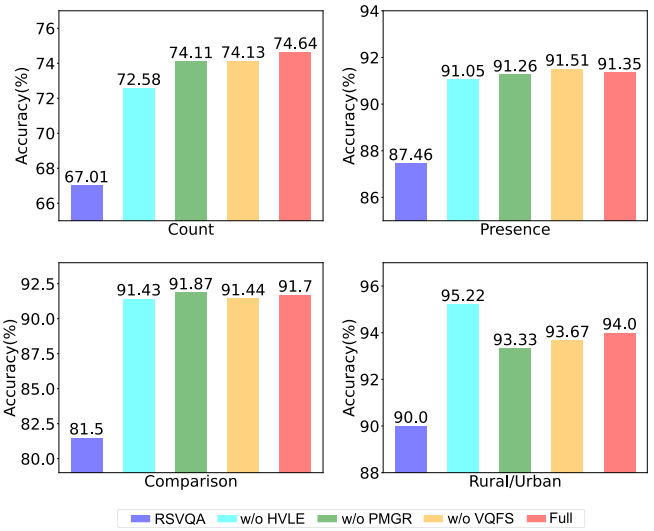


Fig. 7. Visualization of ablation study. The accuracy of different question types on the LR dataset is presented.

compared with the HMGR without the PMGR, the OA and AA of the full model increase by 0.12% and 0.28% on the

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TABLE IV
COMPARISONS PERFORMANCE [MEAN(STD)] OF DIFFERENT BATCH SIZES. THE BEST RESULTS ARE IN BOLD

Batch Size	RSVQA-LR					
	Count	Presence	Comparison	Rural/Urban	Average Accuracy	Overall Accuracy
16	74.29% (0.03%)	91.26% (0.04%)	91.59% (0.02%)	93.33% (0.06%)	87.62% (0.09%)	86.41% (0.07%)
32	**74.64% (0.12%)**	91.35% (0.24%)	**91.70% (0.01%)**	94.00% (0.04%)	87.92% (0.11%)	**86.59% (0.17%)**
64	74.27% (0.02%)	**91.48% (0.06%)**	91.22% (0.07%)	**95.33% (0.11%)**	**88.08% (0.12%)**	86.35% (0.13%)
128	74.20% (0.07%)	91.46% (0.08%)	91.31% (0.12%)	94.67% (0.10%)	87.99% (0.12%)	86.48% (0.09%)

TABLE V
COMPARISONS PERFORMANCE [MEAN(STD)] OF VARIOUS AMOUNT OF TRAINING SAMPLES. THE BEST RESULTS ARE IN BOLD

Utilization Rate	RSVQA-LR					
	Count	Presence	Comparison	Rural/Urban	Average Accuracy	Overall Accuracy
25% training	68.26% (0.05%)	89.09% (0.02%)	90.05% (0.09%)	88.00% (0.64%)	84.05% (0.07%)	83.52% (0.04%)
50% training	73.51% (0.03%)	91.02% (0.07%)	90.84% (0.08%)	**95.33% (0.05%)**	86.24% (0.07%)	85.23% (0.02%)
75% training	72.87% (0.03%)	90.26% (0.13%)	**91.93% (0.15%)**	93.78% (0.21%)	87.21% (0.10%)	85.82% (0.17%)
100% training	**74.64% (0.12%)**	**91.35% (0.24%)**	91.70% (0.01%)	94.00% (0.04%)	**87.92% (0.11%)**	**86.59% (0.17%)**

TABLE VI
COMPARISONS PERFORMANCE [MEAN(STD)] OF NUMBER OF ATTENTION HEADS. THE BEST RESULTS ARE IN BOLD

Head Number	RSVQA-LR					
	Count	Presence	Comparison	Rural/Urban	Average Accuracy	Overall Accuracy
1	73.98% (0.09%)	90.89% (0.13%)	91.02% (0.23%)	93.58% (0.11%)	87.36% (0.08%)	85.83% (0.27%)
2	74.21% (0.16%)	90.96% (0.25%)	91.26% (0.10%)	93.74% (0.15%)	87.54% (0.03%)	86.01% (0.03%)
4	**74.64% (0.12%)**	91.35% (0.24%)	91.70% (0.01%)	**94.00% (0.04%)**	**87.92% (0.11%)**	86.59% (0.17%)
8	74.56% (0.02%)	**91.46% (0.10%)**	**91.81% (0.04%)**	93.82% (0.09%)	87.91% (0.14%)	**86.64% (0.02%)**

TABLE VII
COMPARISONS OF EFFECT OF FINE-TUNING RESNET ON THE THREE DATASETS. THE BEST RESULTS ARE IN BOLD

Variant	RSVQA-LR		RSVQA-HRv1		RSVQA-HRv2	
	OA	AA	OA	AA	OA	AA
frozen ResNet	85.51% (0.10%)	86.40% (0.12%)	84.74% (0.04%)	84.92% (0.13%)	84.98% (0.21%)	84.72% (0.06%)
fine-tuned ResNet	**86.59% (0.07%)**	**87.92% (0.06%)**	**85.53% (0.11%)**	**85.21% (0.02%)**	**85.64% (0.03%)**	**85.15% (0.12%)**

TABLE VIII
COMPARISON ON THE NUMBER OF PARAMETERS AND EXECUTIVE TIME ON THE RSVQA-LR DATASET

Methods	parameters(M)	training time(s)	testing time(s)
RSVQA	85.69	197	34
SHRN	105.56	746	112
HMGR(ours)	97.12	496	54

LR dataset, by 0.22% and 0.23% on the HR(v1) dataset and 0.24% and 0.21% on the HR(v2) dataset. On the other hand, compared to the base model [19], the full model and its three variants show significant performance improvements. By analyzing the quantitative results in Table IV, we can observe that the HMGR model outperforms other variants and significantly improves performance. Besides, as shown in Fig. 7, we visualize the accuracy of different question types on the LR datasets. We can observe two fundamental facts: 1) while the complete model is surpassed by other variants in some types of problems, the margin is not substantial and 2) the complete model and its three variants significantly outperform the base model across various problem categories. Table X shows the incremental ablation manner. By adding the designed modules into the baseline algorithm, all three modules are witnessed to have a significant positive impact on model performance compared to the baseline model. Specifically, the PMGR module shows the greatest improvement, while the VQFS module shows the least improvement.

2) Fine-Tuning of ResNet34: ResNet34 is an effective model designed for visual object classification tasks of natural images. However, images in remoting sensing and natural scenarios exhibit significant differences, especially in imaging viewpoints, imaging resolution, object characteristics, and so on. To fully leverage the feature learning capabilities of ResNet34 for remote sensing images, it is necessary to fine-tune the model on remote sensing datasets, updating its original parameters to achieve domain adaptation. The result of fine-tuning ResNet34 in our method is shown in Table VII. According to the experimental results, we can see a significant

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TABLE IX
ABLATION STUDY ON THE LR, HR(V1), AND HR(V2) DATASETS. THE BEST RESULTS ARE IN BOLD

Variant	RSVQA-LR		RSVQA-HRv1		RSVQA-HRv2	
	OA	AA	OA	AA	OA	AA
Baseline	79.08% (0.20%)	81.49% (0.49%)	83.23% (0.02%)	83.12% (0.03%)	78.23% (0.25%)	77.50% (0.29%)
- HVLE	85.68% (0.04%)	87.57% (0.09%)	85.34% (0.03%)	84.99% (0.11%)	85.43% (0.12%)	84.87% (0.08%)
- PMGR	86.47% (0.06%)	87.64% (0.04%)	85.31% (0.13%)	84.98% (0.12%)	85.46% (0.21%)	84.95% (0.19%)
- VQFS	86.38% (0.08%)	87.69% (0.12%)	85.26% (0.05%)	84.98% (0.10%)	85.51% (0.11%)	84.97 % (0.13%)
HMGR(full)	**86.59% (0.17%)**	**87.92% (0.11%)**	**85.53% (0.21%)**	**85.21%(0.10%)**	**85.64% (0.03%)**	**85.15% (0.12%)**

TABLE X
ADDITIVE ABLATION STUDY ON THE LR DATASETS. THE BEST RESULTS ARE IN BOLD, AND THE SECOND BEST ARE UNDERLINED

Variant	RSVQA-LR	
	OA	AA
Baseline	79.08%(0.20%)	81.49%(0.49%)
+ HVLE	84.53%(0.12%)	84.68%(0.22%)
+ PMGR	84.77%(0.09%)	85.02%(0.14%)
+ VQFS	83.13%(0.13%)	83.75%(0.31%)
HMGR(full)	**86.59% (0.17%)**	**87.92% (0.11%)**

improvement in model performance through the fine-tuning of ResNet34.

E. Time Complexity Analysis

In the experiment, we analyze the computational complexity of two kernel modules, including the multimodal reasoning and the modal fusion. The executive time contains one-epoch training and testing time. Our proposed method is compared to the SOTA method SHRN as reported in Table VIII. For the modal reasoning module, we design a low-rank adaptation graph attention module, which incorporates two important parameter optimization techniques low-rank decomposition and weight sharing to effectively reduce the training parameter size for parameter efficiency. In contrast, SHRN directly employs the classic scaled dot-product attention, resulting in relatively higher computational costs. In the modal fusion stage, we employ pooling operations to aggregate information for feature dimensionality reduction and only utilize one bilinear fusion and one linear mapping. The SHRN method does not perform feature dimensionality reduction and uses linear mapping multiple times, introducing more parameters.

V. CONCLUSION

In this article, we propose the hierarchical multimodality graph reasoning model named HGRM to improve the ability of the RSVQA model toward complicated questions and images. We design the HVLE, the parallel multimodal graph reasoning technique, and the multigranular feature fusion scheme specifically. The representation and interaction of multimodal information is all-sided enhanced. Our approach proves to be effective through comprehensive experiments conducted on three large-scale RSVQA datasets. Nevertheless, our study takes no account of the position and the changes in visual information, leading to a lack of temporal and spatial semantic encoding in the vision–language model. In future endeavors, we plan to devise a temporal–spatial representation and implement a prompt-engineering approach, leveraging pretrained large language models, to further extend the range of answerable questions in the RSVQA task.

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